From Fields to Dose: Review Guide

Evaluating Large Language Models for Weight-Based Pediatric Medication Extraction and Dosing Consistency

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What this package is

This is the review set for the feature article and its supporting documents, issued after four rounds of external adversarial review. It contains the publishable article, the study report that every number in the article traces to, the system architecture document, the six figures as separate files, and this guide. The evidence packet, the code, the locked data and every raw and scored output are in the public repository named below, so any count in the documents can be reproduced from the committed scripts. All five registered models have complete results; the final analysis is in the documents and the figures.

Contents

File	What it is	Read it for
Fields_to_Dose_Feature_Article.docx and .pdf	The publishable text, four sections, one author mention, four figures, sources, editorial notes	The piece under review
Fields_to_Dose_Study_Report.docx and .pdf	Abstract, design, validation, results to date, pending work, deviations, decisions, anticipated challenges, fact-check packet plan, figure map	Every number, rule and decision behind the article
Fields_to_Dose_System_Architecture.docx and .pdf	Repository layout, data model, pipeline, integrity controls, operations, release plan	How the lock, scorer and analysis work in code
Figures/F2D_Figure_Study_Pipeline.png	Figure 1	
Figures/F2D_Figure_Two_Workflows.png	Figure 2	
Figures/ F2D_Figure_Primary_Endpoint.png	Figure 3	five models
Figures/ F2D_Figure_Small_Model_Failure_Composition.png	Figure 4	
Figures/ F2D_Figure_Heatmap_Workflow_B.png	Figure 5 (report only, supporting)	five models
Figures/ F2D_Figure_Heatmap_Workflow_C.png	Figure 6 (report only)	five models

Figure numbers are the same in the article, the report and the evidence packet. The article carries Figures 1 to 4.

Evidence packet (results/packet in the public repository, 31 files)	
01_lock_verification.txt	LOCK.json, shasum of the three data files, hash_scoring() output, empty post-lock commit list for the hashed files, HEAD
02_row_level/	Note, scenario, every model's output and scored row in both workflows for the five stale-weight records, the five maximum-dose notes and the neonatal record
03_written_amount/	admin_amount, by_stratum, per_drug and max_check CSVs, their scripts, the written-amount versus composite cross-tab
04_h3/	Locked and corrected primary analyses, pairwise contrasts, H3

	under both protocol wordings
05_manifest/	Per-file model, endpoint, workflow, run, attempts, first and last timestamps, prompt hash; Ollama digest; models.yaml (environment-variable names only)
06_deviations/	DEVIATIONS.md, LOCK_hashes.txt

Review history

First review round, findings and their resolution:

Finding	Resolution	Where
Seventh frontier error unexplained	Traced to one neonatal record whose printed amount is 13 percent from the exact product; no output can meet both locked tolerances; scored as registered, disclosed, removed by the 20 percent sensitivity	Report 3.2, 5.11
Six maximum-dose outputs not identified	Six-row table: four notes, six outputs, maxima and their fields, label set ids, each model's written amount; the maximum was extracted into the correct field in all six	Report 5.10; article, second finding
Ceftriaxone maximum wording	"1874 mg per administration on a twice-daily order, 3748 mg a day, against a 2 g daily maximum"	Article; Report 5.10
"Changed the kind of failure rather than the amount"	Count rose from 196 to 271; composition stated	Article, third finding
28 and 27 written-amount figures without a table	Named exploratory tool, rule set, denominator, per-level table, missing split by withheld versus printed	Report 5.12
Closing quote merged two causes; "not catchable" too absolute	Stale weight kept as a specification-gap hypothesis; maximum failures named as model behaviour; completeness versus range checks distinguished	Article, closing quote and third finding
Composite versus Delphi threshold	"Composite reconstruction failure" with two components reported separately; "more than 10 percent off" reserved for computed exposures	Throughout

Second review round:

Finding	Resolution	Where
Small model: 196 composite failures versus 54 written-amount problems	Scorer path quoted: workflow B scores the model's reported mg/kg fields; cross-tab shows 154 records with a correct written amount and a failed composite (74 milligram figure in the mg/kg field, 48 per-day value in the per-dose field, 15 null, 17 other); the article says which number a user of each workflow would see and calls the composite a regimen-reconstruction failure	Report 3.4, 5.6; article, third finding; Figure 2
Label claims overstated	Ceftriaxone 2 g per day is the label's; amoxicillin 875 mg and prednisolone 60 mg are the author's documented derivations; the article calls all four "stated in the order"; no pharmacist review is stated as	Report 5.10, 9

	the standing limitation	
"Except the seven already described"	Eight written-amount deviations and seven composite failures reported as separate counts with the difference named	Article; Report 5.12
H3 has two protocol wordings	Both reported side by side with the passages verbatim, paired clustered-bootstrap intervals for each, the reason Section 1 is the analysed definition, and the Section 16 verdict stated as pending the fifth model	Report 5.2; article, third finding
Figure 2 overflow; figure numbering inconsistent	Figure 2 redrawn; one figure map across all documents	This guide; Report 12
OSF registration not independently opened	The fact-check packet carries a dated capture of the registration page and the hash outputs; the lock verification is described step by step, including a discrepancy found and resolved on 5 October	Report 4.3
Closing quote: "not a better prompt"	Removed; the explicit-instruction rerun is listed as the next test	Article

Third review round:

Finding	Resolution	Where
Lead implies the registered measure tests the written amount	Lede and Figure 4 call it the structured-regimen measure; the written-amount result stands beside it	Article
"No answer can satisfy both tolerances" not established	Restated as "no single internally consistent per-dose amount"; a constructed inconsistent output was run through the locked scorer and passed; denominators given (13 percent above the printed amount; 12 percent below the regimen's mg/kg)	Report 3.2, 5.11; article
"A check catches it today"	The check was built and run after the fact over all 2,432 outputs against the model's own extracted maxima: it flags the known outputs, nothing else from the frontier models, and one additional Claude output 7.75 percent over a maximum that the 10 percent tolerance had passed; now seven outputs over five notes; quote rewritten	Report 5.10, 5.14; article
H3 framing	"Under Section 1" and "under Section 16" throughout; the analysed definition is dated by the registration hash, since the analysis script is one of the seven hashed files; tally stated individually with H4 provisional and H5 inconclusive	Report 5.2; article
"Pinned identifiers"	"Identifiers as used"; DeepSeek resolves to its current release; run timestamps and the Ollama digest in the manifest	Report 3.3; packet 05
Eight versus seven accounting	Three-part reconciliation	Report 5.12; article
Heat map too dense for the article; editorial note inconsistent	Heat maps moved to the report as supporting material; article carries Figures 1 to 4; one figure map	Article editorial note; this guide
Evidence packet absent	Built from the repository and published with it	Public repository, results/packet

Fourth review round:

Finding	Resolution	Where
Maximum check stated inconsistently between article and report	One statement everywhere: 2,432 eligible first-run outputs of the four completed models; seven end-to-end and four calculator-workflow flags among the three frontier models; one small-model flag against its own wrong maximum; retrospective, not a tested safeguard	Article; Report 5.14
Figure 4 legend and Figure 2 could be read as testing the written amount	Legends now say reported (B) or reconstructed (C) mg/kg exposure; written-amount result printed on Figure 4	Figures 2 and 4
Evidence packet not accessible	Public repository carries results/packet with the raw and scored files it was built from	Public repository
"Carry beyond this benchmark"; "no two differed"; "five pinned models"; architecture reference	Reworded as patterns within the benchmark; no significant pairwise difference detected, not equivalence; identifiers as used; references aligned	Article; Report abstract, 3.3, 5.3
Development version labels in public documents	Removed; revision history lives in the deviations log and the repository	All documents

Final results with the fifth model

Gemini 3.1 Pro completed all runs on 6 October (run 1 on 361 records in both workflows; three runs on the 108-record subset; zero errors in the final chains), across two Google Cloud projects under one billing account after the per-project daily quota. The final scoring and analysis were run on the full set of 30 files. What changed and what did not:

Item	Final value
Gemini 3.1 Pro, end-to-end	field accuracy 1.000; complete regimen 0.997; composite failure 0.016 (5 of 304); primary endpoint 0.980 (0.964 to 0.993)
Gemini 3.1 Pro, hybrid	all 1.000; composite failure 0
Gemini stability	1.000 (hybrid), 0.991 (end-to-end)
Pairwise	no significant difference among the four frontier models in either workflow (largest adjusted p 0.92 is the smallest); every frontier model exceeds Llama by 0.641 to 0.661 (end-to-end) and 0.941 to 0.944 (hybrid)
Abstention	all four frontier models miss the same five stale-weight records in both workflows, writing the prior weight as current
Maximum-dose outputs	nine over five notes (Claude 5, DeepSeek 2, Gemini 2); seven beyond the tolerance; amoxicillin per-dose maximum missed by three of four frontier models; GPT-5.5 applied the maximum in all five
New: reporting-convention finding	Gemini capped the written amount correctly in four notes and reported the order's nominal mg/kg rather than the capped figure; the prompt does not state the convention; scored as registered (four composite failures with correct written amounts) and disclosed as a clarification
H3 under Section 1 (composite rate)	Claude -0.016, Gemini -0.016 supported; GPT-5.5 null; DeepSeek -0.007 not; Llama +0.247 reversed; 2 of 5, not supported
H3 under Section 16 (primary endpoint)	GPT-5.5 +0.030, Claude +0.016, Gemini +0.020 supported; DeepSeek +0.010 not; Llama -0.273 reversed; 3 of 5, supported
Frontier composite failures, end-to-end	twelve: Claude 5, DeepSeek 2, Gemini 5; all in paraphrased notes
Written-amount deviations, end-to-end	ten: GPT-5.5 1, Claude 4, DeepSeek 3, Gemini 2

Maximum check denominator	3,040 eligible first-run outputs; nine end-to-end and four hybrid flags among the frontier models; one small-model flag against its own wrong maximum
Figures	3, 5 and 6 regenerated with the Gemini row

Still pending: unassisted manual coding and the re-code; TRIPOD-LLM checklist; dated registration capture; approval of any quotation; public repository refresh from the final export; key rotation at acceptance.

What is still pending, and is said to be pending in the documents

1. Public repository refreshed from the final five-model export.

2. The author's written approval of the three quotations; they are drafts.

3. Unassisted manual error coding and the 10 percent re-coding check.

4. The evidence packet refreshed from the final export, plus the dated OSF capture and approved quotes (Report Section 11).

5. The TRIPOD-LLM checklist, released with the paper.

A note on workflow letters

The protocol registered three workflows: a rules-only Workflow A with no language model, which was not run (recorded in the deviations log, with the reason that it lies outside the hypotheses and that the calculator in the hybrid workflow already provides the non-model arithmetic reference), and Workflows B and C, which keep their registered letters because the letters are fixed in the registration, in the locked prompt files and in every result file; renumbering them after registration would make the results disagree with the registered design. The article uses the names end-to-end and hybrid; the report and the files use B and C.

Standing constraints the reviewer should know

Synthetic data only; no patient records. No clinician or pharmacist review; ground truth is by construction and the limitation is stated in the protocol. Solo, self-funded; no institutional resources. All five models pre-registered; none added or dropped. Scorer, calculator, prompts, schema and registered analysis script hashed at lock and verified; every post-registration change is a dated row in the deviations log.

Questions the author would most value an answer to

1. Does the article now state the small model's result in a way that cannot be read as a wrong-dose rate?

2. Is the handling of the two H3 wordings transparent enough, given that the verdict differs between them and one is pending?

3. Is the "stated in the order" treatment of the derived maxima acceptable without a pharmacist read of the five notes, given the stated limitation?

4. Are the three draft quotations consistent with what the evidence supports, as a basis for the author's approval?

5. With the heat maps now in supporting material, do Figures 3 and 4 carry the piece on their own, and is the seventh, within-tolerance output stated at the right weight?